Design and Implementation of an Ultrasonic Sonar Positioning System Using ESP32-Based Master–Slave Architecture

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- **ABSTRACT:**

This project presents the design, modeling, and experimental validation of an ultrasonic sonar-based positioning system using an ESP32-based master–slave architecture. The proposed system operates on the principle of ultrasonic time-of-flight (ToF) measurement, where an ultrasonic pulse is transmitted from a slave node and the reflected echo is received at a master node to estimate distance. Wireless synchronization between the master and slave nodes is achieved using ESP-NOW communication to ensure accurate timing coordination. Two ultrasonic receivers placed at known positions allow the system to estimate the two-dimensional position of the slave node using geometric circle intersection. Mathematical modeling and experimental measurements are conducted to evaluate system accuracy, repeatability, and limitations. Experimental results demonstrate a low overall mean percentage error of approximately 0.63%, indicating reliable short-range indoor positioning performance. The system is low-cost, simple to implement, and suitable for embedded and indoor positioning applications. Environmental effects, synchronization delay, and signal attenuation are also discussed to highlight practical constraints and future improvement scopes.

1. INTRODUCTION:

1.1 Background of Study and Motivation

Ultrasonic sonar systems are widely used in embedded and robotic applications due to their low cost, simplicity, and effectiveness for short-range distance measurement. These systems estimate distance by measuring the travel time of ultrasonic waves through air. With the advancement of wireless microcontrollers such as the ESP32, distributed sensing and synchronized measurement systems have become easier to implement. This project is motivated by the need for an accurate, low-cost, and wireless indoor positioning system that can be implemented using readily available embedded components.

1.2 Project Objectives

The main objectives of this project are:

- To design and implement an ESP32-based master–slave ultrasonic sonar positioning system.
- To develop a time-of-flight-based distance measurement method using ultrasonic sensors.

- To estimate the position of a slave node using two distance measurements.
- To experimentally evaluate system accuracy and reliability under indoor conditions.

1.3 Outline of the Report

This report is organized as follows: Section 2 reviews relevant literature on ultrasonic positioning and ESP32-based systems. Section 3 describes the methodology, modeling, and working principles of the proposed system. Section 4 explains the experimental setup. Section 5 presents the results, discussion, and cost analysis. Section 6 concludes the report and suggests future work.

2.Literature Review

Ultrasonic positioning systems have been extensively studied for indoor localization due to their cost-effectiveness and accuracy at short ranges. Hazas et al. presented a comprehensive survey of ultrasonic positioning systems, highlighting time-of-flight measurement as a reliable distance estimation technique. Liu and Wang demonstrated that ultrasonic ToF-based indoor localization can achieve centimeter-level accuracy under controlled environments, although temperature variation affects accuracy. Kim et al. discussed ESP32-based wireless sensor systems and emphasized their suitability for synchronized, real-time embedded applications. Roy et al. analyzed practical limitations of ultrasonic ranging systems and identified alignment and signal attenuation as major error sources. Zhang et al. proposed a distributed ultrasonic localization system with wireless synchronization and reported improved scalability but noted synchronization latency as a challenge. These studies form the foundation of the proposed ESP32-based master–slave ultrasonic positioning system.

3. Methodology and Modeling:

3.1 Introduction

The proposed system uses ultrasonic time-of-flight measurement combined with wireless synchronization to estimate distance and position. A master–slave architecture is adopted to simplify control and timing management.

3.2 Working Principle of the Proposed Project

The master ESP32 sends a trigger command to the slave ESP32 using ESP-NOW wireless communication. Upon receiving the command, the slave transmits an ultrasonic pulse. The pulse is received by two ultrasonic sensors connected to the master. The travel time of the ultrasonic signal is measured, and distance is calculated using the time-of-flight equation:$d = v \times t$

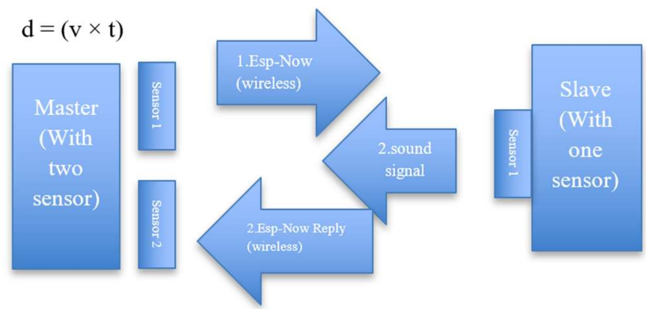

Fig. 1. System block diagram of the ESP32-based ultrasonic sonar positioning system.

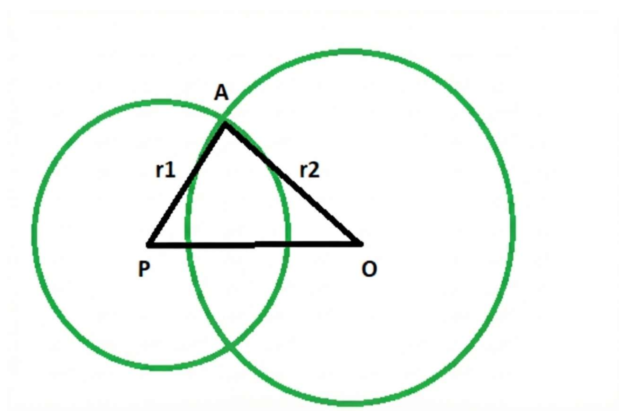

Fig. 2. Graphing the position of slave.
TABLE 1. System Components and Specifications

A= Position
r1, r2=Distance of slave from two sensor
PO= Distance

3.3 Process of Work

The project workflow includes system design, component selection, ESP-NOW communication setup, ultrasonic signal measurement, mathematical modeling, and experimental validation. No external simulation software was used; numerical analysis was performed using real measurement data.

3.4 Description of the Components

The proposed ultrasonic sonar positioning system is composed of several hardware components that work together to perform wireless synchronization, ultrasonic signal transmission, reception, and distance calculation. Each component plays a critical role in ensuring system accuracy, reliability, and low-cost implementation.

ESP32 Microcontroller

The ESP32 is the core processing unit used in both the master and slave nodes. It is a dual-core microcontroller operating at up to 240 MHz and features integrated Wi-Fi and Bluetooth capabilities. In this project, the ESP32 enables wireless communication between nodes using the ESP-NOW protocol, which provides low-latency and peer-to-peer data exchange without requiring a traditional Wi-Fi network. The ESP32 also offers high-resolution timers that are essential for precise time-of-flight measurement of ultrasonic signals. Its low power consumption, affordability, and strong processing capability make it highly suitable for embedded positioning systems.

Fig. 3. ESP32 Microcontroller

Ultrasonic Sensor (HC-SR04)
The HC-SR04 ultrasonic sensor is used for both transmitting and receiving ultrasonic signals at a frequency of 40 kHz. In the slave node, the sensor functions as an ultrasonic transmitter, emitting short bursts of sound waves upon receiving a trigger command from the master. In the master node, two HC-SR04 sensors act as receivers to detect the reflected ultrasonic waves. The sensor measures distance by calculating the time interval between transmission and echo reception. HC-SR04 sensors are widely used due to their low cost, ease of interfacing, and acceptable accuracy for short-range indoor applications.

Fig. 4. Ultrasonic Sensor (HC-SR04)

Power Supply Unit
A regulated 5V power supply is used to power the ESP32 modules and ultrasonic sensors. Stable voltage supply is essential to maintain consistent sensor performance and prevent timing inaccuracies caused by voltage fluctuations. Proper grounding and voltage regulation help reduce electrical noise, which can otherwise affect ultrasonic signal detection and microcontroller operation.

Fig. 5. Power Supply Unit

ESP-NOW Wireless Communication Protocol
ESP-NOW is a lightweight, connectionless wireless communication protocol developed by Espressif. It allows ESP32 devices to exchange data directly with minimal delay. In this project, ESP-NOW is used to synchronize the master and slave nodes, ensuring that ultrasonic pulse transmission occurs immediately after receiving the trigger signal. This tight synchronization is crucial for accurate time-of-flight calculation. Compared to traditional Wi-Fi communication, ESP-NOW offers faster response time and reduced overhead, which significantly improves timing precision.

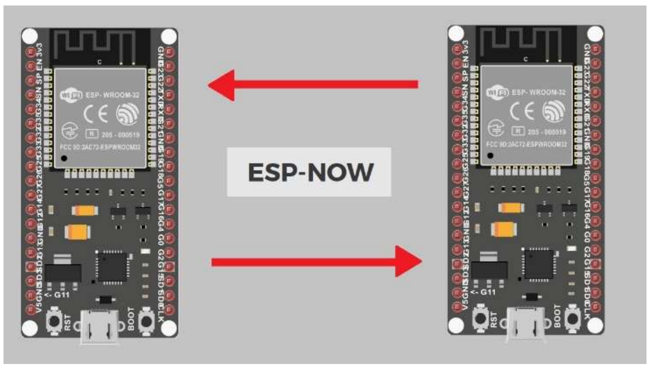

Fig. 6. ESP-NOW Wireless Communication Protocol.

3.5 Test / Experimental Setup

The experimental setup consists of one ESP32 master with two ultrasonic receivers and one ESP32 slave with one ultrasonic transmitter. Measurements are performed at fixed indoor distances to ensure repeatability and reduced environmental interference.

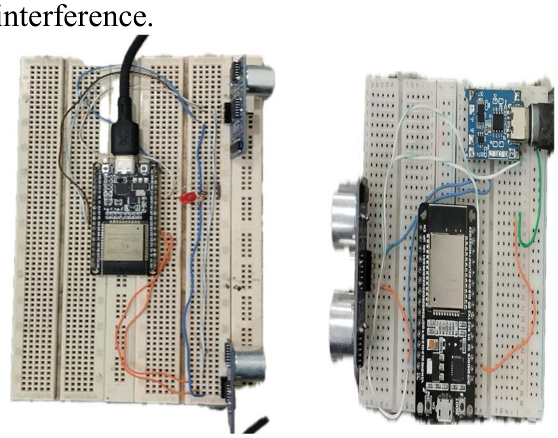

Fig. 7. Experimental setup Master(left) Slave (right)

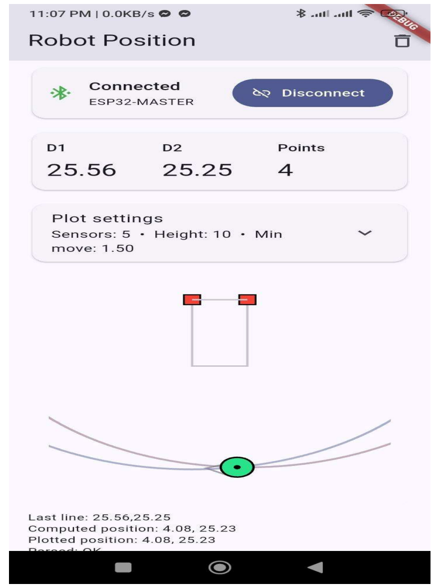

Fig. 8. Showing position with an apps.

4. Results and Discussion

4.1 Numerical and Experimental Analysis

Experimental distance measurements were compared with actual reference distances for two datasets (D1 and D2). Mean absolute errors of 0.76 inch and 0.83 inch were observed for D1 and D2 respectively.

4.2 Comparison of Results

TABLE 1. NUMERICAL ANALYSIS OF REAL DATA AND EXPERIMENTAL DATA

No	D1(Distance Sensor 1) Inch			D2(Distance Sensor 2) Inch		
	Experimental	Actual	Error (%)	Experimental	Actual	Error (%)
1	177.2	179.85	1.47%	129.38	130.97	1.21%
2	68.85	68.62	0.34%	106.32	106.65	0.31%
3	133.45	131.17	1.74%	136.98	139.09	1.52%
4	166.52	166.27	0.15%	106.64	107.52	0.82%
5	97.69	98.03	0.35%	174.31	173.29	0.59%
6	82.46	82.09	0.45%	168.95	169.49	0.32%
7	86.22	87.71	1.70%	106.29	108.11	1.68%
8	168.98	168.98	0.00%	11.96	11.96	0.00%
9	31.64	31.64	0.00%	105.42	105.42	0.00%
10	194.69	194.69	0.00%	112.22	112.22	0.00%

Mean Absolute Error for D1: 0.76 Inch
Mean Absolute Error for D2: 0.83 Inch

Mean Error (%) for D1: 0.62%
Mean Error (%) for D2: 0.64%

Overall error of the system 0.63%

These results indicate good agreement between experimental and actual values.

4.3 Cost Analysis

Component	Quantity	Approx. Cost (BDT)
ESP32	2	Low
HC-SR04 Ultrasonic Sensor	3	Low
Power Supply	2	Low
Total Cost		**Low-cost system**

4.4 Limitations

- Performance is affected by temperature and environmental noise.
- Accuracy decreases with longer distances.
- Wireless synchronization delay introduces small timing errors.

5. Conclusion

This project has successfully presented the complete design, development, and experimental evaluation of an ESP32-based ultrasonic sonar positioning system using a master–slave architecture. The system addresses the need for a low-cost, accurate, and wireless indoor positioning solution using readily available embedded components. By utilizing ultrasonic time-of-flight (ToF) measurement and ESP-NOW-based wireless synchronization, the proposed system accurately estimates distances and determines the position of a slave node using geometric modeling.

Future work will focus on multi-node localization, under water positioning, temperature compensation, and real-time filtering techniques to further enhance accuracy.

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